Force and Laws of Motion

The big idea

Why does a ball keep rolling, and why do you lurch forward when a bus brakes? **Newton's three laws** explain how **forces** change motion.

Balanced and unbalanced forces

A **force** is a push or pull. Forces that cancel out are **balanced**: no change in motion. An **unbalanced** force changes the speed or direction of an object.

Newton's first law (law of inertia)

FIRST LAW

An object remains at rest or in uniform motion in a straight line unless an **unbalanced external force** acts on it.

Inertia is the tendency to resist a change in the state of motion. **Mass is the measure of inertia**: a loaded truck is harder to start or stop than a bicycle.

Examples: passengers fall forward when a bus stops suddenly; dust leaves a carpet when it is beaten; a coin on a card drops into a glass when the card is flicked away.

Newton's second law

Momentum p = mv (unit kg m/s). It has the direction of velocity.

SECOND LAW

The rate of change of momentum is proportional to the applied unbalanced force, in the direction of the force: **F = ma**. The unit of force is the **newton**: 1 N = 1 kg m/s².

A 1000 KG CAR SLOWS FROM 20 M/S TO REST IN 4 S. FIND THE FORCE.

$a = (0 - 20)/4 = -5$ m/s². $F = 1000 \times (-5) =$ **−5000 N** (a 5000 N force opposing motion).

This law explains why a cricketer **pulls his hands back** when catching: more time to stop the ball means less force on the hands. Seat belts and airbags work the same way.

Newton's third law

THIRD LAW

To every action there is an **equal and opposite reaction**. The two forces act on **different bodies**, so they don't cancel.

Examples: a gun recoils when fired; you push the ground back to walk forward; a boat moves back when a sailor jumps forward; a rocket pushes gases down and is pushed up.

Conservation of momentum

LAW OF CONSERVATION OF MOMENTUM

When no external force acts, the **total momentum** of a system stays constant: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$.

A 4 KG GUN FIRES A 20 G BULLET AT 400 M/S. FIND THE RECOIL VELOCITY.

Total momentum before = 0. So $4v + 0.02 \times 400 = 0 \Rightarrow v = \mathbf{-2}$ **m/s** (backwards).

Quick recap

- First law: inertia; mass measures inertia.
- Second law: F = ma; force = rate of change of momentum.
- Third law: equal and opposite forces on different bodies.
- Momentum is conserved without external forces.

Practice questions

1. Inertia of an object depends on its
   (a) speed (b) mass (c) shape (d) colour

2. The SI unit of momentum is
   (a) N (b) kg m/s (c) kg m/s² (d) J

3. A force of 10 N acts on a 2 kg mass. The acceleration is
   (a) 20 m/s² (b) 5 m/s² (c) 0.2 m/s² (d) 12 m/s²

4. Action and reaction forces
   (a) act on the same body (b) act on different bodies (c) cancel each other (d) are unequal

5. A fielder pulls his hands back while catching a ball to
   (a) increase the force (b) increase the time and reduce the force (c) reduce the time (d) stop the ball faster

Answers: 1. b 2. b 3. b 4. b 5. b